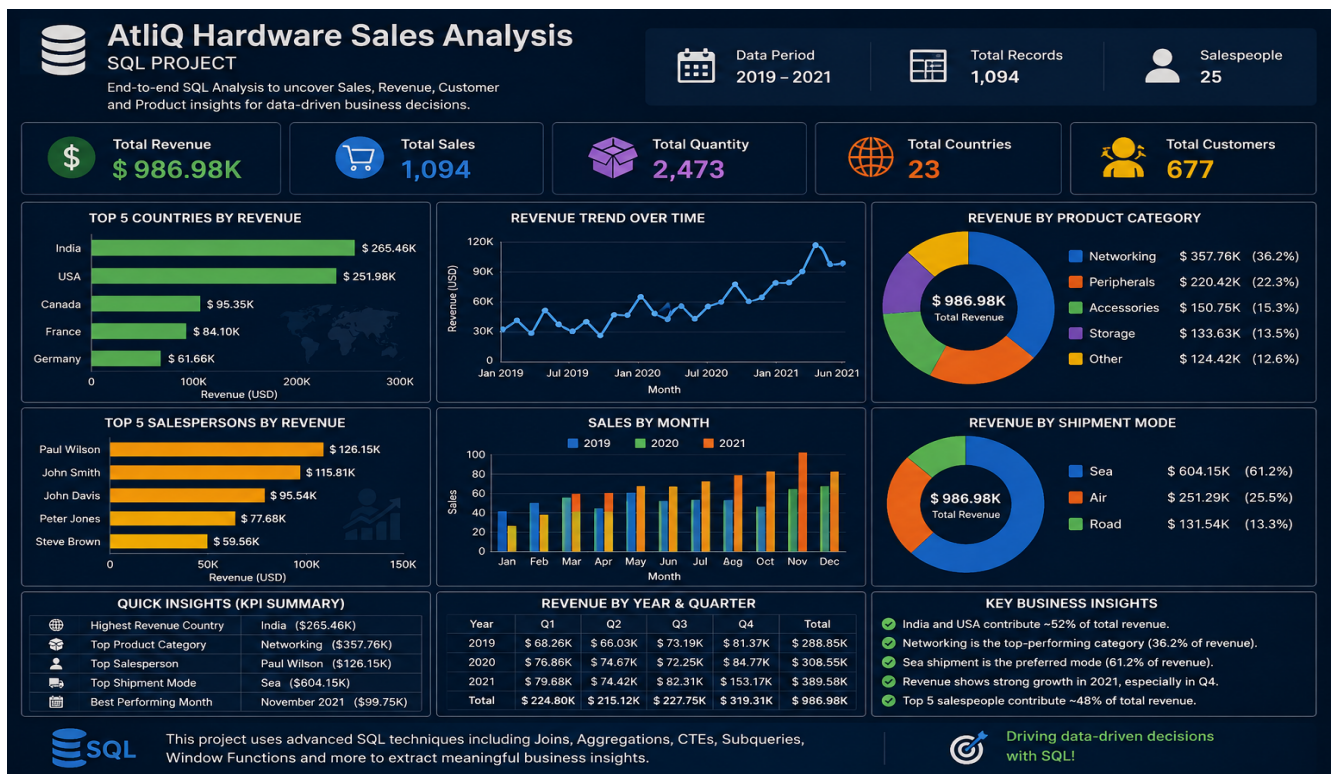


ATLIQ HARDWARE SALES ANALYSIS

SQL PROJECT • SALES PERFORMANCE & REVENUE OPTIMIZATION

An end-to-end SQL analytics project that transforms transactional chocolate sales data into actionable business insights using KPIs, aggregations, CTEs, subqueries and window functions.



Repository-ready documentation

Designed to help a reviewer understand the business problem, database structure, SQL workflow, analysis phases, findings and how the project can be extended.

1. Executive Summary

This project analyzes chocolate sales transactions from a sales-manager perspective. The database contains one central fact table, **ChocSales**, with sales-person, country, product, date, revenue amount and boxes shipped. The SQL work is intentionally organized into four analytical layers: KPI analysis, univariate analysis, bivariate analysis and multivariate analysis.

The purpose is not only to calculate totals, but to explain where revenue comes from, who drives it, which products and markets perform best, how sales volume relates to value, and which combinations of salesperson + country + product deserve management attention.

Project Snapshot

Metric	Value	Metric	Value
Total Revenue	\$6,183,625.00	Transactions	1,094
Boxes Shipped	177,007	Salespeople	25
Countries	6	Products	22
Average Transaction	\$5,652.31	Revenue / Box	\$34.93
Highest Transaction	\$22,050.00	Lowest Transaction	\$7.00
Data Period	03 Jan 2022 – 31 Aug 2022	Best Month	Month 1

Headline findings: Australia is the largest revenue market at \$1,137,367.00 (18.4% of total revenue); Ches Bonnell is the top salesperson at \$320,901.00; Smooth Sliky Salty is the top product at \$349,692.00; and month 1 is the strongest month at \$896,105.00.

Important data note: The supplied dashboard image is a visual reference, but its displayed figures do not match the current Database.sql / Choc Sales dataset (for example, the dashboard shows 2019–2021 and 23 countries, while the supplied database contains 2022 transactions and 6 countries). This report therefore uses the actual database structure and dataset supplied with the project, rather than reproducing the dashboard's different figures.

2. Business Problem & Objectives

Business Problem

A sales organization needs a reliable way to understand revenue performance across people, products, markets and time. Raw transactions alone make it difficult to see concentration, efficiency and performance differences. The project addresses this by turning transaction-level data into decision-ready summaries.

Objectives

Objective	Management Question
Measure overall performance	How much revenue, volume and activity did the business generate?
Compare salespeople	Who generates the most revenue, transactions and shipment volume?
Evaluate markets	Which countries contribute most and how broad is market coverage?
Evaluate products	Which products drive revenue and distribution?
Understand time patterns	Which months and quarters are strongest?
Measure efficiency	How much revenue is generated per box or transaction?
Find high-value combinations	Which salesperson–country–product combinations are strongest?

End Consumer: Sales Manager

The SQL file explicitly frames the project around **Sales Performance & Revenue Optimization** for a sales manager. The analysis is therefore focused on performance ranking, market coverage, product mix, revenue contribution and operational efficiency rather than technical database administration.

3. Database Structure

The supplied Database.sql creates a Microsoft SQL Server database named **ChocSalesDB** and a single table **dbo.ChocSales**. The table is designed as a compact transactional sales fact table.

Column	Type	Purpose
Sales_Person	NVARCHAR(100)	Salesperson responsible for the transaction
Country	NVARCHAR(100)	Market / country where the sale occurred
Product	NVARCHAR(150)	Chocolate product sold
Date	DATE	Transaction date
Amount	DECIMAL(12,2)	Revenue value of the transaction
Boxes_Shipped	INT	Number of boxes shipped

Data Characteristics

The supplied database contains **1,094 transactions**, **25 salespeople**, **6 countries** and **22 products**. The recorded period runs from **03 Jan 2022** to **31 Aug 2022**.

Logical Model

Dimension	Used by SQL for
Salesperson	Revenue ranking, transaction volume, product mix, market reach, productivity
Country	Revenue ranking, market coverage, product availability, contribution analysis
Product	Revenue ranking, transaction mix, market reach, efficiency
Date	Monthly and quarterly trends, month-over-month growth
Amount	Revenue, averages, value bands, contribution percentages
Boxes Shipped	Shipment volume, revenue-per-box and volume bands

4. SQL Methodology

The query file is deliberately staged from simple business KPIs to deeper multi-dimensional analysis. This makes the project easy to explain during a portfolio review or interview.

Technique	How it is used
SELECT / WHERE / ORDER BY	Retrieve, filter and rank result sets
GROUP BY + Aggregations	SUM, AVG, COUNT, MIN and MAX for performance summaries
CTEs (WITH)	Break complex calculations into readable intermediate result sets
Subqueries	Compare grouped metrics with maxima or overall totals
CASE	Create revenue and shipment-volume bands
Window Functions	ROW_NUMBER and LAG for within-group ranking and time comparisons
PARTITION BY	Rank or compare records separately by country
Date Functions	MONTH and DATEPART(QUARTER) for time analysis
Derived Metrics	Revenue per box, revenue per transaction and contribution percentages

Analysis Pipeline

Raw transactions → Database table → KPI layer → Segmentation → Relationship analysis → Ranking & contribution → Business insights

This progression is useful because each phase answers a different class of business question. Phase 1 establishes the baseline; Phase 2 describes individual dimensions; Phase 3 studies relationships between two dimensions; and Phase 4 combines three or more dimensions to reveal stronger patterns.

5. Phase 1 — KPI Analysis

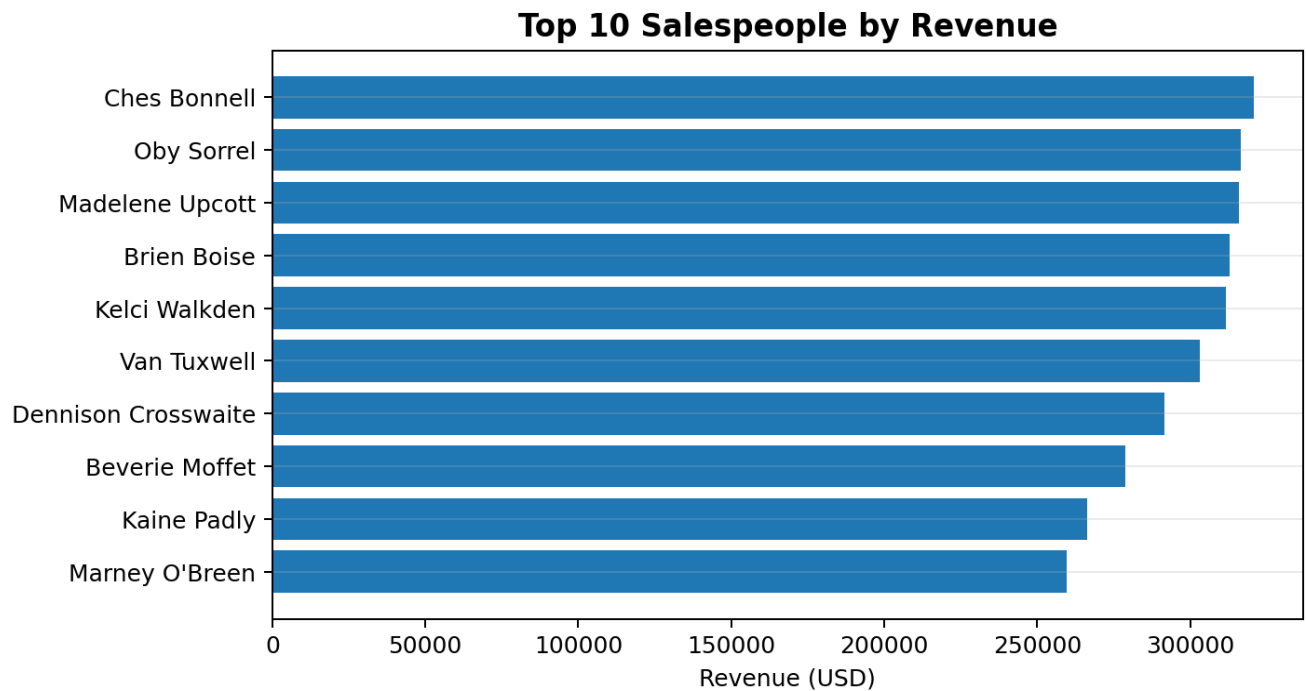
Phase 1 establishes the executive baseline using 15 KPI queries. These queries cover total revenue, shipment volume, transaction activity, efficiency, extremes and top-performing entities.

KPI	Result	Why it matters
Total Revenue	\$6,183,625.00	Overall commercial output
Total Boxes Shipped	177,007	Physical sales volume
Total Transactions	1,094	Sales activity
Average Transaction Value	\$5,652.31	Typical transaction size
Average Boxes / Transaction	161.80	Typical shipment volume
Revenue / Box	\$34.93	Shipment-level efficiency
Highest Transaction	\$22,050.00	Largest individual sale
Lowest Transaction	\$7.00	Smallest individual sale
Salespeople	25	Sales-force size
Countries	6	Market footprint
Products	22	Portfolio breadth
Top Salesperson	Ches Bonnell — \$320,901.00	Highest individual salesperson revenue
Top Product	Smooth Sliky Salty — \$349,692.00	Highest product revenue
Top Country	Australia — \$1,137,367.00	Highest market revenue
Top Country Contribution	18.4%	Revenue concentration

Interpretation: The KPI layer is intentionally simple. It provides the figures that a manager would place on an executive dashboard before drilling into the reasons behind performance.

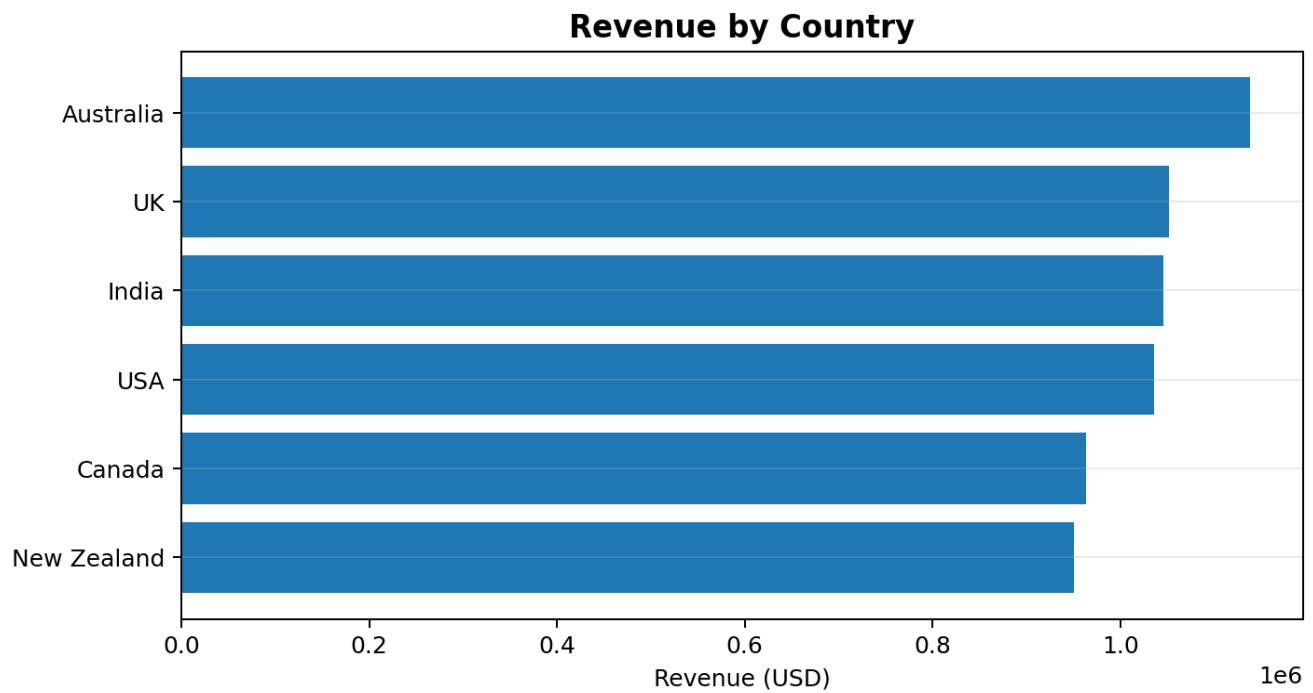
6. Phase 2 — Univariate Analysis

Phase 2 analyzes one business dimension at a time. The query file contains 15 analyses covering salesperson, country, product, month, quarter and distribution bands.



Rank	Salesperson	Revenue	Share
1	Ches Bonnell	\$320,901.00	5.2%
2	Oby Sorrel	\$316,645.00	5.1%
3	Madelene Upcott	\$316,099.00	5.1%
4	Brien Boise	\$312,816.00	5.1%
5	Kelci Walkden	\$311,710.00	5.0%
6	Van Tuxwell	\$303,149.00	4.9%
7	Dennison Crosswaite	\$291,669.00	4.7%
8	Beverie Moffet	\$278,922.00	4.5%
9	Kaine Padly	\$266,490.00	4.3%
10	Marney O'Brien	\$259,742.00	4.2%

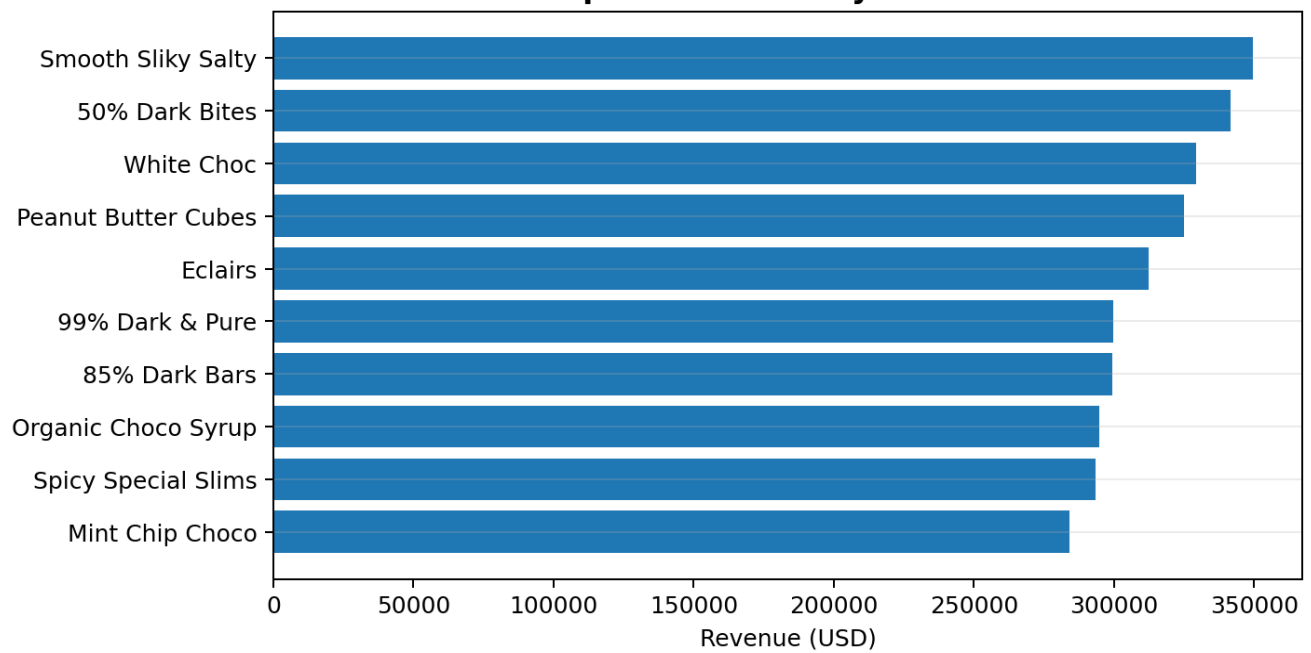
Country & Product Performance



Country	Revenue	Share
Australia	\$1,137,367.00	18.4%
UK	\$1,051,792.00	17.0%
India	\$1,045,800.00	16.9%
USA	\$1,035,349.00	16.7%
Canada	\$962,899.00	15.6%
New Zealand	\$950,418.00	15.4%

Top Products

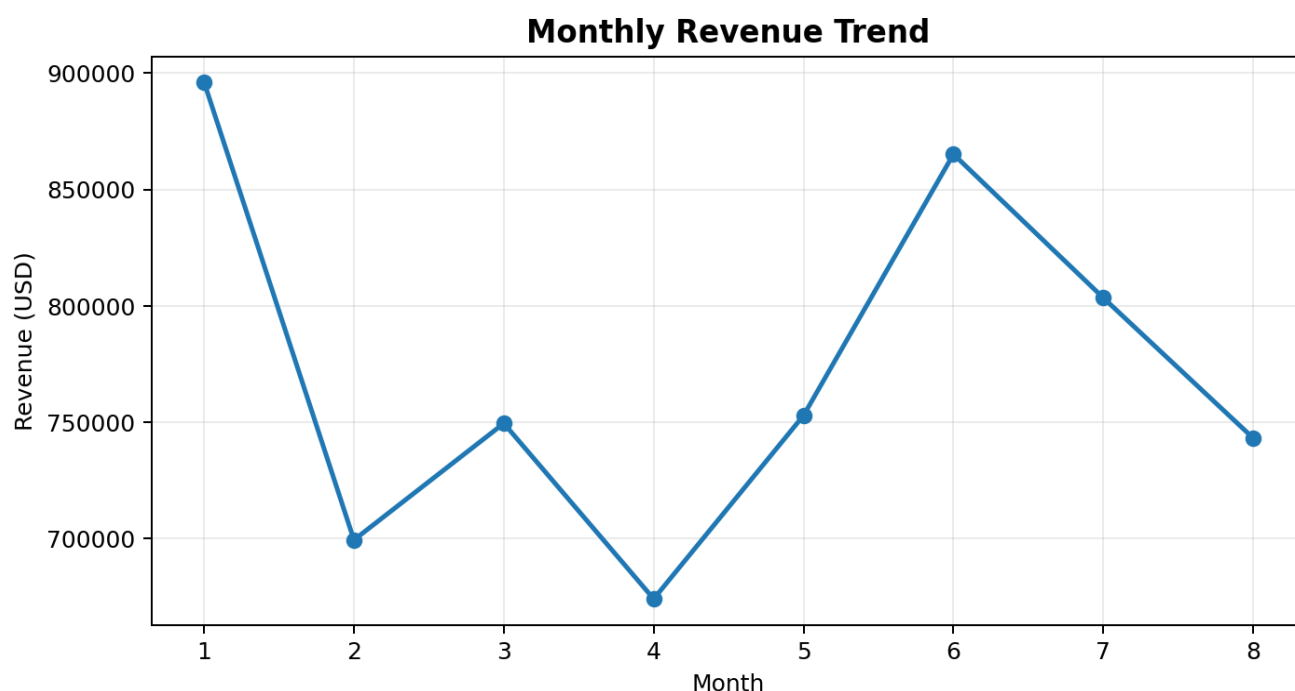
Top 10 Products by Revenue



Product	Revenue	Share
Smooth Sliky Salty	\$349,692.00	5.7%
50% Dark Bites	\$341,712.00	5.5%
White Choc	\$329,147.00	5.3%
Peanut Butter Cubes	\$324,842.00	5.3%
Eclairs	\$312,445.00	5.1%
99% Dark & Pure	\$299,796.00	4.8%
85% Dark Bars	\$299,229.00	4.8%
Organic Choco Syrup	\$294,700.00	4.8%
Spicy Special Slims	\$293,454.00	4.7%
Mint Chip Choco	\$283,969.00	4.6%

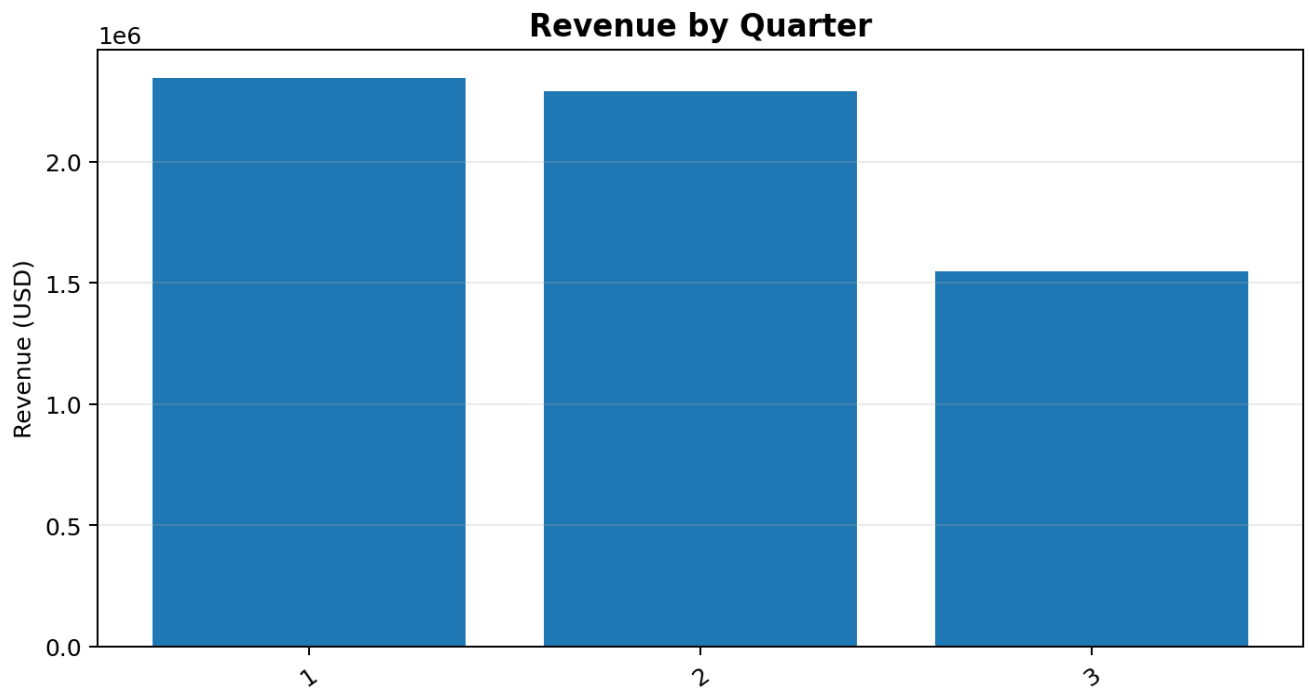
7. Time Analysis — Phase 2

The query set extracts month and quarter from the transaction date to identify seasonality and changes in sales activity. Because the supplied dataset covers January–August 2022, monthly and quarterly comparisons should be interpreted within that available period.



Month	Revenue	Share
1	\$896,105.00	14.5%
2	\$699,377.00	11.3%
3	\$749,483.00	12.1%
4	\$674,051.00	10.9%
5	\$752,892.00	12.2%
6	\$865,144.00	14.0%
7	\$803,425.00	13.0%
8	\$743,148.00	12.0%

Strongest month: Month 1 with \$896,105.00. **Weakest month:** Month 4 with \$674,051.00.



8. Phase 3 — Bivariate Analysis

Phase 3 studies relationships between two dimensions. Instead of asking only 'who sells the most?', the queries ask how shipment volume relates to transaction value, how broad a salesperson's product mix is, how widely products are distributed, and how transaction values are distributed across entities.

Analysis	Business interpretation
Transaction Value vs Box Volume	Compare average transaction value across shipment-volume bands.
Revenue Band vs Box Volume	See whether higher-value sales also involve larger shipments.
Salesperson vs Product Mix	Measure how many distinct products each salesperson sells.
Salesperson vs Market Reach	Measure how many countries each salesperson serves.
Product vs Market Reach	Identify products with broad international distribution.
Salesperson vs Product Diversity	Compare product variety against total transaction activity.
Country vs Product Diversity	Measure portfolio breadth in each market.
Country vs Salesperson Coverage	Measure sales-force coverage by country.
Product vs Salesperson Coverage	Measure how many salespeople sell each product.
Product vs Average Boxes	Identify products associated with larger average shipments.
Country vs Average Boxes	Identify markets with larger typical shipment sizes.
Salesperson vs Average Boxes	Compare shipment size across salespeople.
Product vs Transaction Value Band	See the value distribution of each product.
Salesperson vs Transaction Value Band	See each salesperson's mix of low/high value sales.
Country vs Transaction Value Band	Compare transaction-value distributions by market.

Example Finding: Shipment Volume vs Transaction Value

Shipment Band	Transactions	Average Transaction
Low	401	\$5,837.28
High	170	\$5,693.26
Very High	64	\$5,619.69
Medium	459	\$5,480.08

This type of analysis helps a manager test whether larger shipments are systematically associated with larger-value sales, rather than assuming that volume and revenue always move together.

9. Phase 4 — Multivariate Analysis

Phase 4 is the strongest analytical layer. It combines three or more variables and uses CTEs and window functions to rank entities within markets and calculate contribution and efficiency metrics.

Analysis	Variables combined	Technique
Top Salesperson in Each Country	Salesperson + Country + Revenue	ROW_NUMBER + PARTITION BY
Top 3 Salespeople in Each Country	Salesperson + Country + Revenue	ROW_NUMBER + PARTITION BY
Top Product for Each Country	Product + Country + Revenue	ROW_NUMBER + PARTITION BY
Top 3 Products for Each Country	Product + Country + Revenue	ROW_NUMBER + PARTITION BY
Salesperson + Product Performance	Salesperson + Product + Revenue	ROW_NUMBER
Revenue per Box	Salesperson + Product + Revenue + Boxes	CTE + derived metric
Product Revenue Contribution	Country + Product + Revenue	Window SUM
Salesperson Revenue Contribution	Country + Salesperson + Revenue	Window SUM
Monthly Country Revenue	Month + Country + Revenue	GROUP BY
Monthly Product Revenue	Month + Product + Revenue	GROUP BY
Month-over-Month Growth	Month + Revenue	LAG
Country MoM Growth	Country + Month + Revenue	LAG + PARTITION BY
Salesperson Productivity	Salesperson + Country + Revenue + Transactions	CTE + ratio
Product-Country Efficiency	Country + Product + Revenue + Boxes	CTE + ratio
Performance Scorecard	Salesperson + Country + Product + Revenue + Boxes	CTE + derived metrics

Top Salesperson by Country

Country	Top Salesperson	Revenue
Australia	Mallorie Waber	\$71,134.00
Canada	Beverie Moffet	\$71,995.00
India	Ches Bonnell	\$98,763.00
New Zealand	Madelene Upcott	\$67,550.00
UK	Madelene Upcott	\$71,330.00
USA	Dennison Crosswaite	\$69,545.00

Top Product by Country

Country	Top Product	Revenue
Australia	50% Dark Bites	\$89,222.00
Canada	Smooth Sliky Salty	\$68,257.00
India	Eclairs	\$79,009.00
New Zealand	Mint Chip Choco	\$86,709.00
UK	Peanut Butter Cubes	\$79,695.00
USA	Raspberry Choco	\$83,524.00

10. Key Business Insights

- **1. Australia leads the market.** Australia generates \$1,137,367.00, the highest revenue among the six countries.
- **2. Revenue is relatively distributed across markets.** The largest country contributes 18.4% of total revenue, so the business is not dependent on one market alone.
- **3. Smooth Sliky Salty is the leading product.** It generates \$349,692.00 and is the strongest product in the current dataset.
- **4. Ches Bonnell is the leading salesperson.** With \$320,901.00 in revenue, this salesperson is the top individual contributor.
- **5. Month 1 is the strongest month.** Revenue reaches \$896,105.00 during the available January–August 2022 period.
- **6. Q1 is the strongest quarter.** It generates \$2,344,965.00, ahead of the other completed/available quarters.
- **7. Efficiency matters.** Revenue-per-box and revenue-per-transaction metrics allow management to distinguish high-volume activity from high-value activity.
- **8. Market-level rankings are more actionable than global rankings.** The Phase 4 queries identify the leading salesperson and product inside each country, supporting localized decisions.
- **9. Contribution analysis reveals concentration.** Window functions calculate each product or salesperson's share of a country's revenue.
- **10. The scorecard creates a drill-down path.** Combining salesperson, country and product makes it possible to locate specific high-performing market combinations.

Management Actions Suggested

Action	Reason
Study the leading country	Identify what product mix and salesperson coverage are driving its performance.
Replicate top-product practices	Compare the leading product's market coverage and shipment economics with lower-ranked products.
Coach using high-performing salesperson combinations	Use Phase 4 scorecards to identify repeatable patterns, not just total revenue.
Track revenue per box	Protect against growth that comes mainly from low-value or inefficient shipment volume.
Extend the dataset over time	A longer history would make seasonality and MoM/YoY trends more robust.

11. Technical Skills Demonstrated

Skill	Demonstrated through
SQL Server	Database creation, table creation and SQL-compatible data loading
Data aggregation	SUM, AVG, COUNT, MIN, MAX and GROUP BY
CTEs	Readable multi-step analytical calculations
Subqueries	Top-value comparisons and total-revenue contribution
Window functions	ROW_NUMBER and LAG for ranking and period comparisons
Partitioning	Country-level rankings and country-specific growth analysis
Conditional logic	CASE-based revenue and shipment-volume bands
Date analysis	MONTH and DATEPART(QUARTER)
Business metrics	Revenue per box, revenue per transaction and contribution percentage
Analytical storytelling	Moving from KPIs to relationships and multivariate insights

How to Reproduce the Project

Step 1: Run **Database.sql** in Microsoft SQL Server to create/load the ChocSalesDB database and ChocSales table.

Step 2: Open **Queries.sql** after the table has been created.

Step 3: Execute Phase 1 first to validate the dataset and baseline KPIs.

Step 4: Execute Phases 2–4 progressively for deeper analysis.

Step 5: Export selected result sets to CSV for reporting/dashboard use.

Repository suggestion: Keep the repository organized into *database/*, *queries/*, *results/*, *dashboard/* and *docs/* folders so a reviewer can reproduce the analysis quickly.

12. Query Catalog

The project contains 60 analytical queries across four phases. The catalog below gives a reviewer a quick map from query number to business purpose.

Query	Business Question / Output
P1-01	Total Revenue
P1-02	Total Boxes Shipped
P1-03	Total Sales Transactions
P1-04	Average Transaction Value
P1-05	Average Boxes per Transaction
P1-06	Revenue per Box
P1-07	Highest Transaction Value
P1-08	Lowest Transaction Value
P1-09	Number of Salespeople
P1-10	Number of Countries
P1-11	Number of Products
P1-12	Top Salesperson Revenue
P1-13	Top Product Revenue
P1-14	Top Country Revenue
P1-15	Top Country Revenue Contribution
P2-01	Revenue by Salesperson
P2-02	Transactions by Salesperson
P2-03	Boxes by Salesperson
P2-04	Revenue by Country
P2-05	Transactions by Country
P2-06	Boxes by Country
P2-07	Revenue by Product
P2-08	Transactions by Product
P2-09	Boxes by Product
P2-10	Monthly Revenue
P2-11	Monthly Transaction Volume
P2-12	Monthly Boxes Shipped
P2-13	Quarterly Revenue
P2-14	Transaction Value Distribution
P2-15	Boxes Shipped Distribution
P3-01	Transaction Value vs Box Volume
P3-02	Revenue Band vs Box Volume
P3-03	Salesperson vs Product Mix
P3-04	Salesperson vs Market Reach
P3-05	Product vs Market Reach
P3-06	Salesperson vs Product Diversity
P3-07	Country vs Product Diversity
P3-08	Country vs Salesperson Coverage
P3-09	Product vs Salesperson Coverage

Query	Business Question / Output
P3-10	Product vs Average Boxes
P3-11	Country vs Average Boxes
P3-12	Salesperson vs Average Boxes
P3-13	Product vs Transaction Value Band
P3-14	Salesperson vs Transaction Value Band
P3-15	Country vs Transaction Value Band
P4-01	Top Salesperson in Each Country
P4-02	Top 3 Salespeople in Each Country
P4-03	Top Product for Each Country
P4-04	Top 3 Products for Each Country
P4-05	Salesperson + Product Performance
P4-06	Salesperson + Product Revenue per Box
P4-07	Country + Product Revenue Contribution
P4-08	Salesperson + Country Revenue Contribution
P4-09	Month + Country Revenue
P4-10	Month + Product Revenue
P4-11	Month-over-Month Revenue Growth
P4-12	Country Month-over-Month Growth
P4-13	Salesperson + Country Productivity
P4-14	Product + Country Revenue Efficiency
P4-15	Salesperson + Country + Product Scorecard

13. Conclusion

This project demonstrates how SQL can move from basic aggregation to business-oriented analytical reasoning. The database provides a clean transactional foundation, while the 60 queries progressively answer questions about performance, distribution, relationships, efficiency and market-level combinations. The strongest portfolio value comes from the combination of technical SQL skills and the ability to translate query outputs into management actions.

For a GitHub reviewer, the project can be understood in one flow: data → database → SQL analysis → results → dashboard → business decisions.